From Sound to Occupancy: Privacy-Aware Audio Sensing for Buildings

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Abstract

Occupancy sensors are widely used to control lighting and HVAC in buildings. Passive infrared (PIR) sensors are among the most common, but they respond to motion and often miss people who sit still, so lights or ventilation are turned down in rooms still in use. Sound can reveal them, but microphones raise privacy concerns. We study a two-stage design: an on-device encoder maps each second of audio to an embedding, and a server-side temporal model turns the embeddings into an occupancy state. No public dataset pairs office audio with occupancy labels, so we test the two stages separately on public data. The encoder learns from synthetic scenes mixing empty-room noise, room impulse responses, office sounds and read speech, and is tested on unseen rooms, recordings and speakers. We evaluate whether each second contains a sound made by a person. On 11 scripted real-office recordings (DCASE 2013) the encoder reaches an AUC of 0.87, against 0.76 for log-mel statistics. We attack each representation with speech reconstruction and recognition, closed-set identification of 40 speakers (chance 2.5%) and sex classification (majority 53%). Whisper recovered at most 1% of the words from one-second representations, against 94% from log-mel frames. Speakers remained identifiable: 94 to 98% for log-mel statistics and 69 to 77% for the encoder. Three changes lowered this while keeping detection: noise-floor normalization with adversarial training (55%), an 8-dimensional embedding (47%) and sending only the 16 class probabilities (38 to 47%). Detection AUC stayed at 0.86 to 0.87, and sex accuracy was 60 to 67%. On HPDmobile, which pairs presence labels with an amplitude envelope of the audio, a causal temporal encoder pretrained with hierarchical contrastive learning matches hand-crafted features (AUC 0.75). With one labelled day per training home it did better (balanced accuracy 0.62 versus 0.57 to 0.59, one seed).

1 Introduction

Occupancy sensors are widely used to control lighting and heating, ventilation and air conditioning (HVAC) in buildings. The energy they save depends on how reliably they detect people [13]. Passive infrared (PIR) sensors are among the most common of these devices [29], and like ultrasonic sensors they respond to motion. Much of office work, however, involves little motion: people read, type, think or listen to a lecture for long stretches while barely moving.

As a result, PIR sensors often fail to detect people who stay still [29]. When an occupant stays still for longer than the sensor's time delay, the room is reported vacant and its lights or ventilation are turned down. Long time delays compensate for this uncertainty, but they keep lights on in rooms that are already empty and reduce the savings [13]. As long as the sensor cannot notice a person who sits still, the control system has to trade comfort against energy.

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Several kinds of sensing have been explored to close this gap. Active ultrasonic systems infer occupancy from how a room responds to emitted chirps [27, 28], and multimodal systems combine cameras, microphones and environmental sensors [15, 34]. Passive acoustic sensing suits still occupants, because people keep making sounds while they sit: they type, click, cough, move their chairs, turn pages and talk. Such sounds have been used to detect occupancy and activities in commercial and residential buildings [11, 25, 36]. Because a microphone can also capture conversations, a parallel line of work has designed audio features that allow speech or activity detection without keeping what was said [20, 21, 38]. Others have estimated crowd size or occupancy from non-speech or deliberately perturbed audio [1, 37].

Three problems remain before these approaches can control lighting and ventilation in offices. First, discarding the words does not make audio private. Common spectral features can be turned back into intelligible speech [18, 26]. Where attacks have been tested, information about the speaker remained. Age and gender could still be inferred after speech was masked [5], and individual speakers could be identified from perturbed audio that hid the words [37]. What an attacker can recover from the data a sensor sends, both the words and the speaker, therefore has to be measured. Second, occupancy is a state with memory. A person working quietly may make a sound only every few minutes. A detector that judges each short window on its own cannot tell such a room from an empty one. Third, the data needed to study the problem are missing. Public datasets with occupancy information come from homes. SINS and its DCASE 2018 subset label daily activities in raw audio, including absence [7, 8]. HPDmobile provides long-term occupancy labels, but only with a processed amplitude envelope [15]. Public office recordings, such as those of DCASE 2013, are labelled with sound events rather than presence [31]. To our knowledge, no public dataset pairs audio recorded in offices with labels of whether the room is occupied.

This paper addresses all three problems using public data only. Its main contribution is an evaluation rather than a new architecture: for each audio representation, we measure both how well it detects people and what an attacker can recover from it. We split the sensor into two stages (Figure 1). A small encoder on the device turns each second of audio into a short vector (32 numbers in most experiments), and only these numbers leave the device. A temporal model on the server reads the stream of vectors and tracks whether the room is occupied, using up to 11 hours of past context. Without office occupancy labels, the two stages have to be evaluated separately. The encoder is evaluated on human activity, that is, on whether a second contains a sound made by a person, since this is the evidence the temporal model accumulates. The encoder is trained on synthetic scenes built from public recordings of empty rooms, room impulse responses, isolated office sounds and read speech. It is then tested on scenes from unseen rooms and on real office recordings. We then attack every representation with speech reconstruction followed by speech recognition, speaker identification and sex classification. The temporal model is pretrained on unlabelled sensor streams with the hierarchical contrastive loss of TS2Vec [39] and evaluated on presence in the homes of HPDmobile.

The attacks separate two kinds of leakage. From every one-second representation we tested, the speech recognizer recovered almost none of the words of the reconstructed utterances, while frame-level features gave almost all of them away. Averaging did not remove the speaker. A common baseline is per-second log-mel statistics, the mean and spread of the energy in each frequency band over a second. In a closed set of 40 speakers, these statistics identified nearly every utterance and our task-trained encoder about three quarters. Removing the room's noise floor from each band and training the encoder against a speaker classifier, shrinking the vector to 8 numbers, or sending only the 16 class probabilities each lowered identification, at best to a little over a third of the utterances, while detection of human activity in real offices stayed within about 0.01 AUC of the unprotected encoder (Section 6.2). The representations thus hide the words from our attacks, but they are not anonymous. In homes, the pretrained temporal model matched hand-crafted features when all labels were used and did better when each training home had a single labelled day.

These results also show what is still missing. No experiment here measures occupancy in an office, speakers can still be identified far above chance, and the two stages have not yet been run together. Section 7 discusses these limits, and Section 8 describes the office recordings we plan to collect to address them.

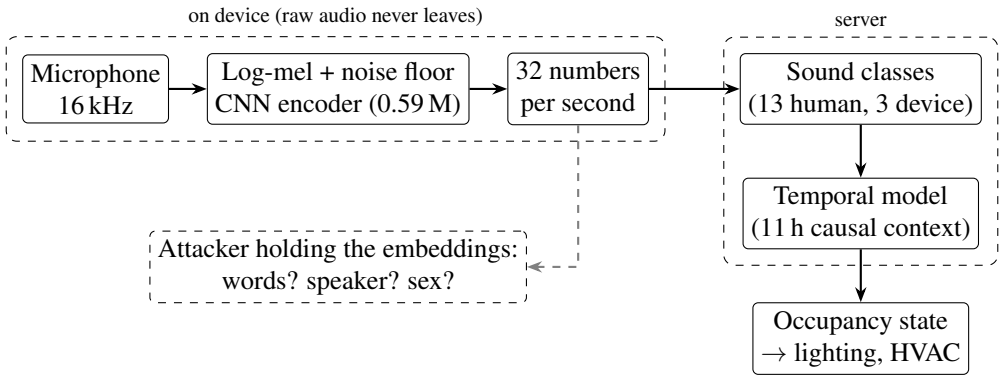


Figure 1: Two-stage design. The device sends one embedding per second; the server detects sound classes and infers an occupancy state from hours of context. The dashed box is the attacker, who holds what the device transmits.

2 Related Work

Occupancy sensing in buildings. Guo et al. reviewed occupancy-based lighting control and noted that current systems detect occupancy at single points with considerable uncertainty [13]. A recent review covers PIR-based people counting, activity detection and localization in buildings [29]. HPDmobile collected images, processed audio and environmental readings with occupancy labels in six homes [15], and the same group later fused these modalities for occupancy detection [34].

Acoustic occupancy and activity sensing. Occupancy has been detected from audio in commercial and residential environments [11], with Gaussian mixtures and hidden Markov models [36], and with engineered audio features [25]. Williams et al. counted overlapping talkers (1 to 10) in simulated cocktail-party mixtures from randomly perturbed audio frames [37]. Crowdotic estimated crowd density in hospital waiting rooms from non-speech audio [1]. Ambient audio has also been proposed for estimating indoor airborne transmission risk factors, including how occupants are distributed [3].

Privacy of audio representations. Beyond the studies discussed in Section 1, Chhaglani et al. described the privacy risks of audio classification and proposed privacy-preserving features [4]. Our attacks follow the approach of Williams et al. [37], but because our representations are not audio, we first reconstruct speech from them before recognition. Adversarial training with a gradient reversal layer [10] is a common way to remove an unwanted attribute from a representation, but the attribute often remains recoverable by a classifier trained afterwards, for demographic attributes in text [9] as for speaker identity in speech representations [30]. At the hardware level, PrivacyMic senses only inaudible ultrasonic frequencies, so that speech is never captured [14]; our approach instead keeps the audible band and limits what leaves the device. Privacy concerns in building data have also motivated learning from anonymized data [6].

Self-supervised representations. Contrastive learning yields general-purpose audio representations [24]. TS2Vec learns timestamp-level representations of time series with a hierarchical contrastive loss over random overlapping crops [39]. We use this loss to pretrain the temporal occupancy model on unlabelled sensor streams.

3 Problem Setup and Threat Model

A sensing device in a room records audio at 16 kHz. For each second t it computes a representation v_t and sends it to a server; raw audio is neither stored nor transmitted. In most experiments v_t is an embedding; in one variant it is the vector of class probabilities computed on the device (Section 6.2). The server outputs a human-activity score s_t per second and an occupancy state $y_t \in \{0,1\}$ for control. For utility, the server should detect human activity, including the quiet sounds of still occupants, in rooms not seen during training. For privacy, v_t should not allow recovery of spoken words and should reveal as little as possible about who is speaking. For deployability, the encoder must be small enough to run on the device.

Human activity and occupancy are different targets. Human activity means that a second contains a sound made by a person, while occupancy means that someone is present, whether or not they make a sound. Our office experiments measure the first and our home experiments the second.

The adversary obtains the transmitted representations, for example through a compromised server, and knows the encoder. In the strongest setting we test, it also holds labelled speech of the target speakers and clean, close-talking recordings. It tries to (a) reconstruct and transcribe speech, (b) identify the speaker among a known set, and (c) infer the speaker's sex. Attacks on the device itself, which has access to raw audio, are out of scope. We give no formal privacy guarantee and report only what these attacks recover.

4 Method

On-device encoder. Each second of audio is converted to 64-band log-mel energies $m_{f,\tau}$ for band f and frame τ (25 ms windows, 10 ms hop), giving a 64×100 patch. Noise-floor normalization (NF) subtracts, in each band, the 10th percentile over the set $\mathscr{W}$ of the 1,000 frames in the 10-s block that contains the current second:

$$\tilde{m}_{f,\tau} = m_{f,\tau} - Q_{0.1}\big(\{m_{f,\tau'} : \tau' \in \mathscr{W}\}\big). \tag{1}$$

The features then describe deviations from the background of the room and microphone. After batch normalization of each band, a convolutional network g_ϕ maps each normalized one-second patch $\tilde{M}_t$ to an embedding $z_t = g_\phi(\tilde{M}_t) \in \mathbb{R}^d$. It has four blocks of two 3×3 convolutions (32, 64, 128 and 128 channels) with batch normalization and ReLU, and 2×2 average pooling between blocks. Its output is averaged over frequency, mean- and max-pooled over time, and projected to d dimensions, with $d = 32$ unless stated otherwise. The network has 0.59 million parameters and sees only the current second. Through Eq. (1), however, the features also depend on the 10-s block, so the device processes audio in 10-s blocks and emits the embeddings of a block once the block is complete, up to 10 s after the sound. When sound fills the whole block, the 10th percentile is no longer the level of the empty room; we did not treat this case separately.

Activity head. On the server, a layer-normalized two-layer perceptron h_ψ with 128 hidden units predicts the probabilities of 16 sound classes, $p_t = \sigma(h_\psi(z_t)) \in [0,1]^{16}$. Thirteen classes are produced by people (clear throat, cough, door slam, drawer, keyboard, keys, knock, laughter, mouse click, page turn, pen drop, speech, light switch) and three by devices (alert, phone ringing, printer). The human-activity score of a second is the largest probability among the human classes $\mathscr{H}$,

$$s_t = \max_{c \in \mathscr{H}} p_{t,c}. \tag{2}$$

Encoder and head are trained with the binary cross-entropy over the $T = 10$ seconds and 16 classes of a scene,

$$\mathscr{L}_{\text{ev}} = -\frac{1}{16T} \sum_{t=1}^{T} \sum_{c=1}^{16} \Big[y_{t,c} \log p_{t,c} + (1 - y_{t,c}) \log(1 - p_{t,c}) \Big], \tag{3}$$

where $y_{t,c} = 1$ if class c is active in second t. Unlike sound event detectors that model context across frames and seconds [2], encoder and head use no temporal context; that is left to the occupancy model.

Speaker-adversarial training. An adversary a_ω, a perceptron with two hidden layers of 256 units, predicts the speaker of a scene among 251 training speakers. Its input is the mean $\bar{z}$ and the element-wise maximum $\hat{z}$ of the embeddings over the speech-active seconds, and its loss for the true speaker k is $\mathscr{L}_{\text{adv}} = -\log a_\omega(k \mid \bar{z}, \hat{z})$. Scenes without speech (60% of them on average) contribute only to $\mathscr{L}_{\text{ev}}$. A gradient reversal layer [10] between encoder and adversary leaves the forward pass unchanged and multiplies the gradient by $-\lambda_p$ on the way back, so that training solves

$$\min_{\omega} \mathscr{L}_{\text{adv}}, \qquad \min_{\phi,\psi} \mathscr{L}_{\text{ev}} - \lambda_p \mathscr{L}_{\text{adv}}. \tag{4}$$

The adversary learns to identify the speaker, while the encoder keeps what the sound classes need and removes what helps the adversary. Following Ganin and Lempitsky [10], the weight grows with training progress $p \in [0,1]$ as

$$\lambda_p = \lambda \left(\frac{2}{1 + e^{-10p}} - 1 \right), \qquad \lambda \in \{1, 3\}. \tag{5}$$

Temporal occupancy model. For long-term occupancy we use a causal dilated convolutional encoder f_θ that maps the inputs up to time t to a representation $r_t \in \mathbb{R}^{64}$. Its receptive field is long because, in the homes of Section 6.3, sound concentrates around arrivals and departures and the occupancy state has to be carried through long quiet stretches. A linear input projection to 64 channels is followed by 10 residual blocks and a 1×1 output convolution to 64 channels. Each block has two convolutions with 64 channels, kernel size 3, dilation 2^i ($i = 0, \ldots, 9$) and GELU activations, padded on the past side only. The receptive field is 4,093 steps, or 11.4 hours at one step per 10 s.

We pretrain f_θ without labels using the hierarchical contrastive loss of TS2Vec [39]. Two overlapping random crops of each sequence are encoded with random timestamp masking, which zeroes each projected input step with probability 0.5, giving representations $r_{i,t}$ and $r'_{i,t}$ of sequence i at each timestamp t in the overlap. The temporal loss contrasts each timestamp with the other timestamps Ω of the same sequence. The instance-wise loss contrasts it with the same timestamp of the other sequences in a batch of size B:

$$\ell^{\text{temp}}_{i,t} = -\log \frac{\exp(r_{i,t} \cdot r'_{i,t})}{\sum_{t' \in \Omega} \left(\exp(r_{i,t} \cdot r'_{i,t'}) + \mathbf{1}[t \neq t'] \exp(r_{i,t} \cdot r_{i,t'}) \right)}, \tag{6}$$

$$\ell^{\text{inst}}_{i,t} = -\log \frac{\exp(r_{i,t} \cdot r'_{i,t})}{\sum_{j=1}^{B} \left(\exp(r_{i,t} \cdot r'_{j,t}) + \mathbf{1}[i \neq j] \exp(r_{i,t} \cdot r_{j,t}) \right)}. \tag{7}$$

Both losses are averaged over sequences, timestamps and the two crops and combined with equal weight. The representations are then max-pooled along time by a factor of 2, and the loss is computed again. This repeats until a single timestamp remains, where only the instance-wise term applies. The pretraining loss is the mean over these levels.

For prediction we fit a class-balanced logistic regression on the frozen representation and the time of day (linear probe),

$$P(y_t = 1) = \sigma\big(w^\top [\, r_t;\ \sin 2\pi\tau_t;\ \cos 2\pi\tau_t \,] + b\big), \tag{8}$$

where $\tau_t \in [0,1)$ is the time of day as a fraction of 24 hours. We also report fine-tuning the pretrained network and training the same network from scratch; in both cases a linear layer on $[\, r_t;\ \sin 2\pi\tau_t;\ \cos 2\pi\tau_t \,]$ is trained together with f_θ.

Pretraining draws windows of 3,000 steps (8.3 h) from the pretraining homes, skipping windows in which more than half of the steps have no data, and runs 600 iterations of AdamW (learning rate 10^{-3}) on

batches of 8 windows; the common part of the two crops has a random length of 2 to 3,000 steps. Fine-tuning and training from scratch use binary cross-entropy with the positive class weighted by the ratio of negative to positive training labels, for 800 iterations of AdamW (learning rate 10^{-3}, weight decay 10^{-2}) on batches of 8 windows of 6,144 steps (17 h), with no loss on the first third of each window, where the causal context is short. The probe is an L_2-regularized logistic regression (regularization strength $C = 1$, class weights inversely proportional to class frequency) on standardized inputs, fitted on every sixth labelled step, that is, one step per minute.

5 Experimental Setup

5.1 Data

Table 1 lists the public datasets. We used only recordings whose licenses allow research use, and we report the license stated by each host.

HPDmobile. Jacoby et al. instrumented six homes in Boulder, Colorado, apartments and single-family houses with one to three or more occupants, for four to eight weeks each in 2019 [15]. Each home had four to six sensor hubs in common areas, never in bedrooms or bathrooms, and each hub recorded monochrome images at one frame per second, 10-s audio clips at 8 kHz, and temperature, humidity, equivalent CO_2, volatile organic compounds and illuminance every 10 s. Raw audio was not released: every clip was mean-shifted, full-wave rectified and downsampled by a factor of 100 before publication, which removes intelligible speech. Occupancy ground truth combines phone geofencing logs with paper sign-in sheets, reconciled by the authors and corrected where the camera showed a person during a period marked vacant, and is given as a home-level occupied or vacant state for every 10-s interval. Audio from three to five hubs per home was available to us.

DCASE 2013 office sounds. The event detection task of the first DCASE challenge [31] covers 16 office sound classes. Its training material is a set of isolated recordings of each class, and its office live (OL) test set consists of 11 recordings of one to three minutes in several office environments, in which people performed scripted, non-overlapping sequences of the 16 sounds; the recordings were annotated by two annotators.

DCASE 2016 Task 2. This task reused 11 of the 16 DCASE 2013 classes, dropping those judged too short or too broad, with new isolated sounds recorded in offices at LS2N in France [17]. The release contains 20 training sounds per class and a public test set of 54 two-minute mixtures that the organizers synthesized from a pool of 40 sounds per class at event-to-background ratios of -6, 0 and $+6$ dB, with monophonic and polyphonic conditions.

LibriSpeech. LibriSpeech is a corpus of read English speech derived from public-domain audiobooks, sampled at 16 kHz and segmented into utterances with transcripts and speaker labels [19]. The train-clean-100, dev-clean and test-clean subsets contain 251, 40 and 40 speakers, with no speaker shared between subsets; the "clean" subsets hold the speakers whose recordings a baseline recognizer transcribed most accurately, and the recordings are close-talking and free of noticeable noise.

BUT ReverbDB. The BUT Speech@FIT Reverb Database was recorded by Brno University of Technology in nine rooms: three offices, two lecture rooms, a meeting room, a conference room, a hotel room and a stairwell [32]. Each room was measured with 31 microphones and two to ten loudspeaker positions, giving

Table 1: Public datasets used in the experiments. For each dataset: the part we use, the labels it provides, the license stated by its host, and its role (sources of office sounds, speech and room acoustics for the synthetic scenes, test sets, speech for the privacy attacks, or input to the temporal model).

Dataset	Content used	Labels	License	Role
HPDmobile [15]	6 homes, 152 labelled days; 10-s clips of an 80 Hz rectified amplitude envelope from 3 to 5 hubs per home	Home occupied, every 10 s	CC0	Temporal model
DCASE 2016 Task 2 [17]	220 isolated office sounds (11 classes); 54 test mixtures of 120 s	Class, onset, offset	CC BY 3.0 / 4.0	Event sources; test set
DCASE 2013 OL/OS [31]	320 isolated office sounds (16 classes); 11 scripted recordings in real offices (1,601 s)	Class, onset, offset (two annotators)	CC BY 4.0	Event sources; real test set
LibriSpeech [19]	train-clean-100 (251 speakers), dev-clean (40), test-clean (40)	Transcript, speaker, sex	CC BY 4.0	Speech sources; attacks
BUT ReverbDB [32]	9 rooms (3 offices, 2 lecture rooms, meeting room, conference room, hotel room, stairs), 31 microphones; empty-room noise (60 s) and impulse responses (1 s)	None	CC BY 4.0	Backgrounds; room acoustics

room impulse responses from exponential sine sweeps and, for every loudspeaker position, a recording of the empty room.

5.2 Office benchmark

Training examples are 10-s scenes synthesized on the GPU. Each scene starts from a random 10-s excerpt of one empty-room recording of one training room. We add 0 to 4 isolated sounds at random positions, with event-to-background ratios drawn from -10 to 15 dB. With probability 0.4 we also add a 2 to 8 s excerpt of one utterance at a signal-to-noise ratio from -5 to 20 dB. Every source is convolved with an impulse response of the same room, drawn independently for each source, so the sources of a scene can sit at different positions; a random gain of ± 10 dB is then applied to the mixture. A second is labelled with a class if any 10-ms frame of the dry source lies within 30 dB of the source's loudest frame. From each room we keep a random subset of at most 300 impulse responses and 24 empty-room recordings across all microphone positions and loudspeaker setups; the empty-room recordings of one BUT setup, in which a radio played in the background, are excluded.

Training uses six rooms: two offices, stairs, a hotel room, a conference room and one lecture room. It also uses the first 15 recordings of each class from each DCASE dataset and 24 utterances from each of the 251 train-clean-100 speakers. The 11 classes of DCASE 2016 are a subset of the 16 classes of DCASE 2013 (its keys-drop class is treated as keys), so both collections feed one 16-class label set. We test on three sets. The first has 1,200 synthetic scenes (12,000 s) from three unseen rooms: office L212, lecture room D105 and meeting room C236. These scenes use the remaining 5 recordings per class and test-clean speech. The second is the 11 real office recordings of DCASE 2013 OL, labelled with the union of both annotators. The third is the 54 DCASE 2016 test mixtures (6,480 s). Human activity occupies 44.7%, 41.7% and 39.4% of the seconds in these sets. Each configuration is trained once, for a fixed 3,000 steps of 48 scenes (about 400 hours of audio), with AdamW (weight decay 10^{-4}), gradient clipping at norm 5 and a one-cycle learning rate up to 2×10^{-3}; there is no validation set and no early stopping.

Labels in all three office test sets mark sound events, not presence, so a quiet second is negative even if someone is in the room. The synthetic test set holds out rooms, individual sound recordings and speakers.

Still, the isolated sounds of a class come from the same DCASE collection and may share recording sessions, performers and equipment. It therefore measures generalization to new rooms and voices more than to new recording setups. The DCASE 2013 OL recordings come from the same challenge as part of our training sounds, so equipment or performers may overlap with training even though no recording does. Whether the pool of 40 sounds per class behind the DCASE 2016 test mixtures overlaps the 20 released training sounds is not documented, so we checked directly. For each of the 1,620 annotated events in the test mixtures, we computed the maximum normalized cross-correlation between the mixture segment and every released training sound of the same class. The median was 0.20 and no event exceeded 0.69, whereas for the development mixtures, which the organizers built from the training sounds, the median was 0.78 and 63% of the events exceeded 0.7. The test mixtures therefore do not contain the training sounds.

5.3 Baselines

We compare the encoder with three fixed representations. Log-mel frames at 100 Hz (64 values) offer no privacy and serve only as a reference for the attacks. Per-second log-mel statistics take the mean and standard deviation of each band (128 values per second), with and without NF. The envelope follows the HPDmobile processing (8 kHz audio, mean shift, full-wave rectification, every 100th sample kept), giving 80 values per second. For fixed representations, the same activity head is trained on the same scenes.

5.4 Metrics

Detection is scored with the human-activity score s_t of Eq. (2) in offices and with the predicted occupancy probability of Eq. (8) in homes, second by second and without temporal smoothing. We report the area under the ROC curve (AUC), which is threshold-free and equals the probability that a random positive second scores above a random negative one (0.5 is chance), and the F1 score at a fixed threshold of 0.5, which also depends on how well the scores are calibrated. Appendix C adds average precision (AP) and the equal error rate (EER), the error at the threshold where the false positive and false negative rates are equal. For homes we add balanced accuracy, the mean recall over the two classes.

5.5 Privacy attacks

All attacks use LibriSpeech speakers that the encoder never heard.

- **Speaker identification:** the utterances of the 40 test-clean speakers are split alternately into 1,319 for training the attacker and 1,301 for testing. The attacker is a two-layer bidirectional gated recurrent unit (GRU) network with 128 units per direction over the standardized representation sequence, with mean and max pooling over time, trained for 40 epochs with AdamW (learning rate 10^{-3}, weight decay 10^{-2}, batches of 32 utterances); for 100 Hz inputs, strided convolutions first reduce the rate to 25 Hz. Chance is 2.5%.

- **Sex classification:** the same architecture is trained for 20 epochs on dev-clean (2,703 utterances, 40 speakers) and tested on test-clean (2,620 utterances, 40 other speakers). The majority class is 53.0%.

- **Speech reconstruction:** a decoder (linear layer, convolution, two-layer bidirectional GRU, 256 units) maps the representation, repeated to 100 Hz, to an 80-band log-mel spectrogram. It is trained for 15 epochs on dev-clean with the same learning rate and weight decay, and tested on 200 test-clean utterances of at most 30 s. Each reconstruction is inverted with 60 iterations of Griffin-Lim [12] and transcribed with Whisper small.en [22]. We report the short-time objective intelligibility (STOI) [33],

which is higher for more intelligible speech, and the share of reference words recovered in order,

$$R = \frac{\sum_u \mathrm{LCS}(w_u, \hat{w}_u)}{\sum_u |w_u|}, \tag{9}$$

where w_u and $\hat{w}_u$ are the reference and recognized word sequences of utterance u and LCS is the length of their longest common subsequence. Unlike the word error rate, R ignores inserted words. This matters because the recognizer invents long transcripts for unintelligible input, which pushes the word error rate above 100%.

5.6 HPDmobile protocol

Because HPDmobile releases only the amplitude envelope, the encoder of Section 4 cannot be applied to it; the temporal model is evaluated on per-slot statistics of the envelope, which stand in for the per-second embeddings of a deployment. For each home and 10-s slot, we summarize every hub's 800-sample clip by the logarithm of one plus each of six statistics: the mean, standard deviation, maximum, and 10th, 50th and 90th percentiles. We take the maximum and the mean of each feature over hubs and add the fraction of hubs reporting and a missing-data flag, giving 14 input channels. Each feature is then standardized per home with that home's median and interquartile range, without labels. Days without data are kept as gaps so that the time axis is continuous; slots without data enter the network as zeros, marked by the flag, and training windows in which more than half of the slots have no data are skipped. We evaluate leave-one-home-out, training on five homes and testing on the sixth, for every home. Pretraining uses either the five training homes (strict) or all six homes without labels (transductive), since a deployed sensor accumulates unlabelled data at its own site. The baselines are class-balanced logistic regression and gradient boosting (scikit-learn defaults otherwise) on hand-crafted features: the 14 inputs above, the time of day and trailing means over 1, 5, 30 and 120 minutes of four of the inputs (the clip mean and clip maximum, each as maximum and as mean over hubs); like the probe, they are fitted on every sixth labelled slot. We report balanced accuracy, F1 of the occupied class at a threshold of 0.5 and AUC, averaged over the six held-out homes. For full-label runs, we average over three seeds. For label efficiency we keep the labels of 1, 3 or 7 random days per training home.

6 Results

6.1 Detecting human activity

Table 2 reports per-second detection of human activity. To judge the differences, we resampled whole test recordings with replacement (2,000 bootstrap draws) and computed 95% intervals. The differences quoted below are means of the paired differences over these draws, so they can deviate in the last digit from a subtraction of the table entries. Appendix B lists the intervals together with results for each synthetic test room. The learned encoder outperforms both fixed representations on all three test sets. The gap is largest on real office recordings, where the encoder with NF exceeds log-mel statistics with NF by 0.109 (95% interval 0.067 to 0.147). NF adds 0.035 (0.010 to 0.059) to the encoder on real offices and doubles its F1 at a fixed threshold, from 0.25 to 0.50. This suggests that much of the remaining error came from differences in recording level between rooms and microphones.

Adversarial training changes the real-office AUC by -0.007 (-0.019 to 0.003) for $\lambda = 1$ and by 0.005 (-0.012 to 0.023) for $\lambda = 3$. Its effect on detection is therefore too small to measure. It does lower the F1 at the fixed threshold, so the scores need recalibration. The 8-dimensional embedding loses 0.010 (0.002 to 0.020) on real offices, and adversarial training at this size loses another 0.120. Across the three synthetic test rooms, the AUC of the encoder with NF ranges from 0.939 to 0.963. The room thus matters more than

Table 2: Per-second detection of human activity (a sound made by a person in that second) on the three office test sets: 1,200 synthetic scenes in three unseen rooms, the 11 real office recordings of DCASE 2013 OL, and the 54 DCASE 2016 test mixtures. AUC and F1 at a threshold of 0.5. NF: noise-floor normalization; adv. λ: speaker-adversarial training with weight λ; d: embedding size (32 unless stated). Without NF, adversarial training was run only with $\lambda = 1$.

	Unseen rooms (synthetic)		Real offices (DCASE 2013)		DCASE 2016 test	
Representation	AUC	F1	AUC	F1	AUC	F1
HPDmobile-style envelope	0.699	0.400	0.621	0.275	0.726	0.263
Log-mel statistics	0.873	0.684	0.682	0.041	0.609	0.159
Log-mel statistics + NF	0.894	0.699	0.760	0.447	0.895	0.460
Encoder	0.959	0.852	0.834	0.251	0.950	0.436
Encoder + adv. 1	0.958	0.751	0.832	0.170	0.935	0.365
Encoder + NF	0.947	0.837	0.869	0.502	0.961	0.753
Encoder + NF + adv. 1	0.949	0.779	0.863	0.354	0.957	0.609
Encoder + NF + adv. 3	0.945	0.698	0.874	0.237	0.964	0.354
Encoder + NF, $d = 8$	0.954	0.830	0.859	0.462	0.966	0.705
Encoder + NF + adv. 3, $d = 8$	0.913	0.687	0.740	0.315	0.928	0.425

the choice among the better configurations. These intervals cover the variation between test recordings, not between training runs.

Sounds other than speech. The still occupant that motivates this work makes no speech, so Appendix C repeats the evaluation on the seconds without speech, with the non-speech human sounds as positives. In the real office recordings, where speech covers only 68 of the 668 positive seconds, the picture is unchanged: the encoder with NF reaches an AUC of 0.862 (0.839 to 0.884), 0.117 (0.074 to 0.155) above log-mel statistics with NF, and the adversarial variants stay within 0.01 of it. On the synthetic sets the non-speech AUC is lower than the pooled AUC (0.908 against 0.946 in unseen rooms for the encoder with NF), because speech is the easiest class to detect and the non-speech sounds are short and often quiet. Average precision and the equal error rate rank the configurations in the same order as the AUC.

6.2 What an attacker can recover

Table 3 and Figure 2 summarize the attacks. Frame-level log-mel features expose everything: the reconstructed speech yields 94.3% of the reference words, the same as Griffin-Lim inversion of the true spectrogram and close to the 94.5% that Whisper recovers from the clean audio, and the speaker is identified 98.2% of the time. Reducing the rate to one vector per second defeats this attack for every representation we tested. At most 1.0% of the reference words are recovered, and STOI falls to 0.37 to 0.44. This shows that our decoder and recognizer fail, not that every attacker would.

Per-second log-mel statistics, however, still identify 97.8% of utterances among 40 speakers, so the speaker survives the lower rate. A likely reason is that a speaker's pitch range and timbre change little within a sentence and survive averaging over a second, whereas words change within a fraction of a second. Classical speaker recognition rests on the same fact: the distribution of short-term spectral features over an utterance identifies the speaker [23]. The task-trained encoder keeps less identity (76.8%). NF lowers it to 68.6%, and adversarial training lowers it further to 60.3% ($\lambda = 1$) and 55.3% ($\lambda = 3$). Sex classification falls from 75.3% to 60.1%, close to the 53.0% majority. The HPDmobile-style envelope reveals the least identity (22.4%) but also detects activity worst.

Shrinking the embedding lowered identification more than adversarial training did. With NF and no

Table 3: Privacy attacks on LibriSpeech test-clean speakers, whom the encoder never heard. Rate × size: vectors per second and their dimension. Words: share of reference words recovered in order by Whisper from the reconstructed speech (Eq. 9); STOI: short-time objective intelligibility of the reconstructed speech, higher being more intelligible. Speaker ID: closed-set identification among 40 speakers (chance 2.5%); Sex: binary classification (majority class 53.0%). Probabilities: only the 16 class probabilities of the activity head are sent, so detection equals that of the same encoder in Table 2.

Representation	Rate × size	Words (%)	STOI	Speaker ID (%)	Sex (%)
Log-mel frames (reference)	100 Hz × 64	94.3	0.96	98.2	94.3
Log-mel statistics	1 Hz × 128	0.8	0.39	97.8	88.6
Log-mel statistics + NF	1 Hz × 128	1.0	0.39	94.2	80.5
HPDmobile-style envelope	1 Hz × 80	0.9	0.44	22.4	55.8
Encoder	1 Hz × 32	0.9	0.39	76.8	81.9
Encoder + adv. 1	1 Hz × 32	0.4	0.39	62.3	82.6
Encoder + NF	1 Hz × 32	0.7	0.39	68.6	75.3
Encoder + NF + adv. 1	1 Hz × 32	0.5	0.38	60.3	73.2
Encoder + NF + adv. 3	1 Hz × 32	0.5	0.39	55.3	60.1
Encoder + NF	1 Hz × 8	0.1	0.38	46.6	66.7
Encoder + NF + adv. 3	1 Hz × 8	0.2	0.39	33.6	85.5
Probabilities, Encoder + NF	1 Hz × 16	0.3	0.38	47.1	65.4
Probabilities, Encoder + NF + adv. 3	1 Hz × 16	0.1	0.37	37.5	60.3

adversary, the 8-dimensional embedding identifies 46.6% of utterances, fewer than any 32-dimensional encoder. Adversarial training at this size lowers identification to 33.6%, but it costs detection and raises sex classification from 66.7% to 85.5%. The adversary is trained to remove speaker identity, not sex, so lowering one attribute need not lower the other.

Sending only the class probabilities lowered identification further without changing detection. In this setting the device transmits the 16 outputs of the activity head instead of the embedding. The server computes the human-activity score of Eq. (2) from these outputs anyway, so detection is identical to that of the same encoder. The attacker identified 47.1% of utterances from the probabilities of the encoder with NF. With adversarial training ($\lambda = 3$), this fell to 37.5%, the lowest value among the settings that keep detection performance. Sex classification reached 65.4% and 60.3%. This costs flexibility: the server sees only the 16 trained classes, and adding a class requires updating the device.

Even in the settings that keep detection performance, the attacker still names the right speaker among 40 in 38 to 55% of the utterances, 15 to 22 times chance. The test favors the attacker, since the speech is clean and close-talking, every candidate is known and the attacker trains on labelled speech of each one. A real deployment would capture far-field speech mixed with room noise, which we have not tested. Until such conditions are measured, the transmitted data should be treated as able to identify speakers. Their protection should not rest on the low word recovery alone.

6.3 Occupancy in homes

Label alignment and the role of transitions. HPDmobile labels mark whether anyone is home, including quiet periods such as nights. Loudness alone predicts home occupancy poorly. Taking the log mean envelope of each slot, its maximum over hubs and a 30-minute moving average, the AUC against the labels is 0.70, 0.52, 0.67, 0.44, 0.60 and 0.39 for the six homes, below chance in two of them. Shifting the labels against this loudness by -12 to $+12$ h in steps of 30 min did not reveal a clock offset. Averaging the per-minute loudness around label changes instead shows that every home is loudest in the 10 minutes after an arrival and before a departure (Figure 3; peak z-scores 0.63 to 1.01). Since occupied periods also include long quiet

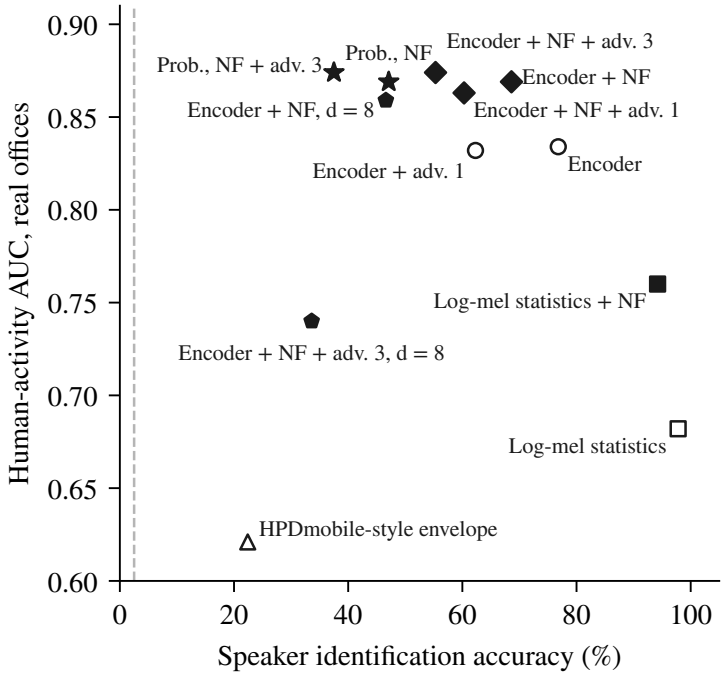


Figure 2: Speaker identification accuracy among 40 speakers (Table 3) against the human-activity AUC on the real office recordings (Table 2), one marker per representation. Filled markers use noise-floor normalization (NF); pentagons are 8-dimensional embeddings; stars (Prob.) send only the 16 class probabilities. The dashed line marks chance identification (2.5%).

nights, occupancy is carried by sparse transition events and must be remembered through silence.

Leave-one-home-out results. Table 4 reports the mean over the six held-out homes, averaged over three seeds. The linear probe on the pretrained temporal encoder matches the best hand-crafted baselines in AUC, with 0.752 under both strict and transductive pretraining against 0.748 and 0.743. Its balanced accuracy is the highest (0.678 and 0.673, against 0.649 and 0.654). Pretraining on the unlabelled audio of the test home did not change these results beyond the variation between seeds. Training the same network end to end, with or without pretraining, generalizes worse to unseen homes (AUC 0.68 to 0.71), which suggests overfitting to the five training homes. The probe did not, however, beat hand-crafted features whose longest trailing mean spans 120 minutes. These results therefore do not show that context beyond two hours was needed; we have not varied the receptive field.

Label efficiency. With fewer labels, the advantage of pretraining grows (Figure 4, one seed). With one labelled day per training home, the probe reaches a balanced accuracy of 0.616, against 0.565 and 0.590 for the hand-crafted baselines. The network trained from scratch reaches only 0.515. In AUC, the probe stays best or tied at 1, 3 and 7 labelled days (0.691, 0.706 and 0.722).

7 Limitations

What was measured. The office experiments measure per-second human activity, not occupancy, and no office test set contains quiet but occupied periods. The non-speech evaluation of Appendix C shows that

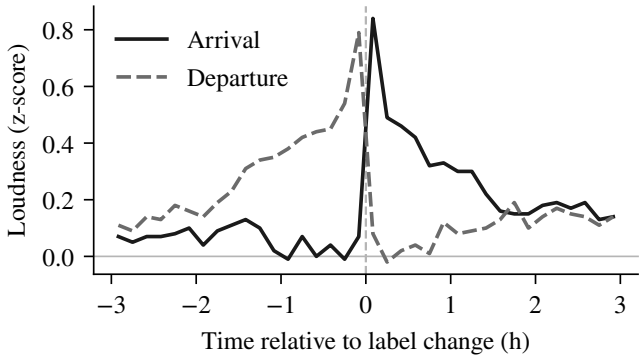


Figure 3: Loudness around changes of the occupancy label in HPDmobile: z-score of the per-minute maximum envelope in 10-min bins, averaged over the six homes, from 3 h before to 3 h after an arrival (solid) or a departure (dashed). Time 0 is the label change.

Table 4: Home occupancy on HPDmobile, leave-one-home-out: balanced accuracy, F1 of the occupied class at a threshold of 0.5, and AUC, averaged over the six held-out homes and over three seeds; subscripts give the standard deviation over seeds. Strict: pretraining on the five training homes; transductive: pretraining also on the unlabelled audio of the test home. The hand-crafted baselines use no pretraining and gave the same result for every seed.

	Strict			Transductive		
Method	Bal. acc.	F1	AUC	Bal. acc.	F1	AUC
Hand-crafted + logistic regression	0.649	0.676	0.748	0.649	0.676	0.748
Hand-crafted + gradient boosting	0.654	0.712	0.743	0.654	0.712	0.743
Same network, labels only	$0.614_{\pm.007}$	$0.713_{\pm.032}$	$0.708_{\pm.008}$	$0.608_{\pm.028}$	$0.706_{\pm.037}$	$0.699_{\pm.007}$
Pretrained + fine-tuning	$0.605_{\pm.018}$	$0.704_{\pm.042}$	$0.684_{\pm.033}$	$0.596_{\pm.018}$	$0.658_{\pm.032}$	$0.686_{\pm.020}$
Pretrained + linear probe	$0.678_{\pm.006}$	$0.716_{\pm.010}$	$0.752_{\pm.005}$	$0.673_{\pm.008}$	$0.697_{\pm.012}$	$0.752_{\pm.003}$

the sounds of a working occupant are detected, not that a silent one is. The training scenes are synthetic, and the only real office test set is small (11 scripted recordings, 1,601 s). It also comes from the same challenge as part of our training sounds. Each office configuration was trained once, so our intervals cover the variation between test recordings but not between training runs. We did not compare the encoder with audio taggers pretrained on large labelled collections, such as PANNs [16], which would show how far training on synthetic scenes is from the detection that larger models reach. The temporal model was trained on homes, with whole-home labels and an amplitude envelope as input. The two stages of Figure 1 have therefore not yet been evaluated together. Its advantage over hand-crafted features was small with all labels and larger with one to seven labelled days, which we ran with a single seed. Deployability was argued from the size of the encoder (0.59 million parameters); we have not measured its latency or power on a target device.

Privacy. Speaker identity was reduced but not removed, and under adversarial training sex classification can move in the opposite direction from identity (Section 6.2). The adversary used during training, a perceptron over pooled embeddings, is weaker than the recurrent attacker we evaluate, so adversarial training may understate what a matched adversary could remove. The attacks also cover only part of the threat: a closed set of 40 speakers, one language and attackers that we trained ourselves, so a stronger attacker could recover more. Closed-set identification is also a different task from the open-set verification with which speaker anonymization is evaluated [35], and a drop in closed-set accuracy need not carry over to verification [30],

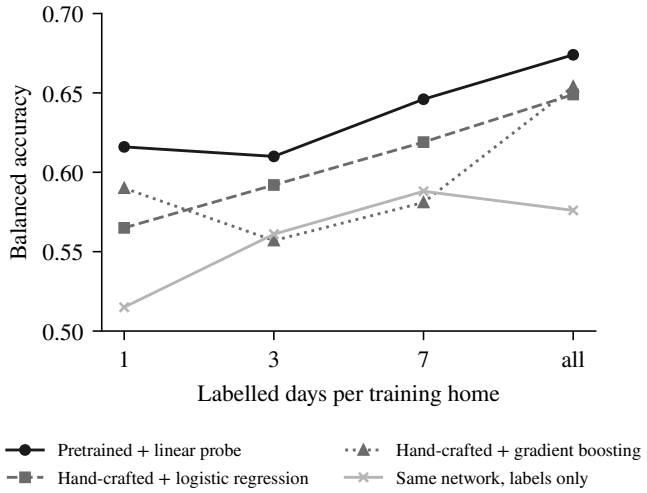


Figure 4: Label efficiency on HPDmobile: balanced accuracy, averaged over the six held-out homes, when each training home keeps the labels of 1, 3 or 7 random days or of all days (transductive pretraining, seed 0).

so our numbers are not comparable with that literature.

Calibration. On real office recordings, the F1 at a fixed threshold ranges from 0.04 to 0.50 across representations and falls with stronger adversarial training, even when the AUC does not. Scores therefore have to be calibrated for each room and model before a threshold can drive control.

8 Future Work

The main next step is an office dataset that we will collect ourselves and report in a separate journal article. We plan to record in single-occupant labs, shared labs and lecture rooms on the University of Illinois campus. Several rooms of each kind will be included, so that whole rooms can be held out for testing. Each room will be recorded for at least two weeks, including nights and weekends to capture the sound of the empty room. The same low-cost device, a single-board computer with a small digital microphone, will be installed in every room next to a commercial PIR sensor. Ground truth will come from entry and exit sign-in sheets and from magnetic door sensors with automatic timestamps. In lecture rooms, class schedules and attendance counts will be used as well. In some sessions, participants will deliberately sit quietly, for example reading for 30 minutes, to test the case that PIR sensors miss. For our sensor and the PIR sensor alike, we will measure how often occupied periods are judged vacant and how long each takes to switch off after people leave. Training the temporal model on these office streams will evaluate the two stages together. On the public benchmarks, the next steps are to repeat the label-efficiency study and the office configurations over several seeds, to pretrain across many buildings, to add a pretrained audio tagger as a reference for detection, to train the adversary with the attacker's architecture, to vary the receptive field of the temporal model, and to test simple temporal smoothing of the per-second scores. Long-term recording will store only the representations computed on the device. Raw audio will be recorded only in short sessions with consenting participants and stored encrypted. The study will begin only after approval by the university's Institutional Review Board.

For privacy, we will add quantization and noise to the transmitted class probabilities and train an adversary for sex next to the one for speaker identity. The attacks will be repeated with far-field speech recorded

in the target rooms, with open-set speaker verification and with other languages. For calibration, we will use empty periods such as nights, which occur at every site and need no labels. The threshold of each room can then be set from the scores observed during those periods. Beyond this paper, we plan to fuse the audio representations with PIR and LiDAR sensing. We also plan to evaluate closed-loop lighting and ventilation control for energy and comfort, and to extend the study to homes.

9 Conclusion

A one-second acoustic representation computed on the device can detect the sounds people make while working, such as typing, page turns and coughs, in rooms it has never heard. Our reconstruction attack recovered almost none of the words from it. Who is speaking, however, survives the reduction in rate and has to be removed deliberately. Noise-floor normalization combined with adversarial training, a smaller embedding or class-probability outputs lowered speaker identification from 77% to 38 to 55%. Detection stayed at a real-office AUC of 0.86 to 0.87, but identification is still far from chance. In homes, occupancy shows up as short loud transitions separated by long quiet stretches. A temporal model pretrained without labels makes the best use of this structure when labels are scarce. Office recordings with occupancy labels are the next step.

Data and Code Availability

All datasets are public (Table 1). Code for data preparation, training, attacks, the bootstrap analysis, the non-speech evaluation and the DCASE 2016 overlap check is available at `https://github.com/AaronChae/occupancy-audio-privacy`.

Ethics Statement

This study uses only public datasets under their stated licenses and collects no new data from people. The planned office recordings will follow the Institutional Review Board process described in Section 8.

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A Per-Home Results

B Bootstrap Intervals and Per-Room Results

Table 6 gives 95% percentile intervals of the human-activity AUC, obtained by resampling whole recordings with replacement (2,000 draws): 11 recordings for real offices and 54 mixtures for DCASE 2016. The last

Table 5: AUC on each held-out home of HPDmobile (strict pretraining, seed 0). The last two rows give the number of labelled days and the share of occupied 10-s slots in each home.

Method	H1	H2	H3	H4	H5	H6
Hand-crafted + logistic regression	0.861	0.698	0.833	0.717	0.760	0.620
Hand-crafted + gradient boosting	0.700	0.745	0.889	0.702	0.806	0.616
Same network, labels only	0.622	0.706	0.871	0.607	0.748	0.639
Pretrained + fine-tuning	0.671	0.578	0.845	0.573	0.795	0.555
Pretrained + linear probe	0.826	0.708	0.815	0.707	0.761	0.666
Labelled days	30	17	35	17	15	38
Occupied share (%)	82.2	60.4	78.2	81.7	75.6	46.8

three columns give the AUC in each unseen synthetic room, computed on 400 new scenes per room, so they differ slightly from the pooled synthetic results in Table 2. Paired differences between configurations, computed on the same draws, are reported in the code repository.

Table 6: Human-activity AUC on the two recorded test sets with 95% bootstrap intervals (2,000 resamplings of whole recordings: 11 for real offices, 54 mixtures for DCASE 2016), and AUC in each unseen synthetic room (400 new scenes per room).

Representation	Real offices	DCASE 2016 test	L212	D105	C236
HPDmobile-style envelope	0.621 (0.587–0.655)	0.726 (0.695–0.757)	0.704	0.694	0.748
Log-mel statistics	0.682 (0.617–0.740)	0.609 (0.587–0.631)	0.869	0.855	0.879
Log-mel statistics + NF	0.760 (0.726–0.794)	0.895 (0.884–0.905)	0.907	0.913	0.885
Encoder	0.834 (0.816–0.854)	0.950 (0.942–0.957)	0.945	0.959	0.953
Encoder + adv. 1	0.832 (0.817–0.847)	0.935 (0.927–0.943)	0.942	0.958	0.946
Encoder + NF	0.869 (0.847–0.892)	0.961 (0.954–0.967)	0.948	0.963	0.939
Encoder + NF + adv. 1	0.863 (0.833–0.888)	0.957 (0.950–0.963)	0.948	0.964	0.930
Encoder + NF + adv. 3	0.874 (0.862–0.887)	0.964 (0.958–0.970)	0.944	0.960	0.929
Encoder + NF, $d = 8$	0.859 (0.831–0.884)	0.966 (0.961–0.972)	0.949	0.965	0.948
Encoder + NF + adv. 3, $d = 8$	0.740 (0.699–0.774)	0.928 (0.920–0.936)	0.919	0.930	0.885

C Detection of Non-Speech Human Sounds

Table 7 restricts the evaluation to the seconds whose labels contain no speech. A second is positive if a non-speech human sound (clear throat, cough, door slam, drawer, keyboard, keys, knock, laughter, mouse click, page turn, pen drop or light switch) is active and negative if no human sound is active; seconds with speech are dropped. This leaves 600 positive and 933 negative seconds in the real offices, 2,127 and 3,930 in the DCASE 2016 test mixtures, and 2,760 and 6,624 in the 1,200 synthetic scenes of Appendix B. The score is the same human-activity score as in Table 2; intervals are 95% bootstrap intervals over whole recordings (2,000 draws), as in Table 6. Per-second results for all seconds, with AP and EER, are in the code repository.

Table 7: Detection of non-speech human sounds: AUC with 95% bootstrap interval, average precision (AP) and equal error rate (EER) on the seconds whose labels contain no speech, with the non-speech human sounds as positives. Intervals as in Table 6.

	Real offices (DCASE 2013)			DCASE 2016 test			Unseen rooms (synthetic)		
Representation	AUC	AP	EER	AUC	AP	EER	AUC	AP	EER
HPDmobile-style envelope	0.622 (0.589–0.655)	0.564	0.418	0.713 (0.683–0.742)	0.657	0.352	0.683	0.467	0.363
Log-mel statistics	0.660 (0.592–0.723)	0.593	0.395	0.542 (0.519–0.566)	0.545	0.479	0.775	0.598	0.280
Log-mel statistics + NF	0.745 (0.709–0.782)	0.673	0.312	0.879 (0.868–0.890)	0.841	0.173	0.832	0.654	0.214
Encoder	0.825 (0.803–0.846)	0.804	0.248	0.941 (0.931–0.949)	0.932	0.106	0.915	0.816	0.143
Encoder + adv. 1	0.824 (0.807–0.838)	0.786	0.260	0.924 (0.915–0.933)	0.894	0.120	0.911	0.775	0.147
Encoder + NF	0.862 (0.839–0.884)	0.839	0.220	0.953 (0.946–0.960)	0.941	0.089	0.908	0.771	0.140
Encoder + NF + adv. 1	0.855 (0.826–0.881)	0.822	0.217	0.949 (0.941–0.957)	0.936	0.100	0.905	0.748	0.143
Encoder + NF + adv. 3	0.869 (0.857–0.882)	0.817	0.200	0.958 (0.950–0.965)	0.942	0.084	0.899	0.734	0.152
Encoder + NF, $d = 8$	0.850 (0.823–0.874)	0.829	0.236	0.960 (0.954–0.966)	0.947	0.089	0.915	0.784	0.134
Encoder + NF + adv. 3, $d = 8$	0.719 (0.679–0.752)	0.595	0.308	0.916 (0.907–0.925)	0.826	0.102	0.845	0.638	0.194